AILA: An Intelligent Lecturing Assistant for High-Quality Formative Assessment Generation

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Problem Statement

Instructors face challenges distilling key concepts and creating formative assessments from diverse lecture materials. Most AI tools do not automate this process reliably.

How Can AI Enhance Efficiency and Quality in Automated Question Generation?

Project Objective

Web platform for uploading lectures, extracting and deduplicating concepts, building knowledge graphs, and generating Multiple Choice Questions (MCQs).

Data Collection & Methodology

- ▶ **Source:** Lecture slides: Drexel Course CS172 OOP
- ▶ **Generation:** AI pipelines (KG, Raw). Specific prompts
- ▶ **Dataset:** 257 KG MCQs, 207 Raw MCQs
- ▶ **Evaluation criteria:** 14 metrics, Likert scale (1–5)

	Raw MCQ	KG-based MCQ (RAG)
Context	Current segment/slide only	All key concepts for entire week/lecture
Input	Raw text of the segment	JSON list of concepts (knowledge graph)
LLM Prompt	"Given this **content**, make MCQs."	"Given these **concepts**, make MCQs based on concepts and their relationships."
Question Focus	**Comprehension**, fact-recall, details in one chunk	**Conceptual**, higher-order, link/mix ideas
Difficulty Style	Straightforward, local to slide	Creative, big-picture, across slides
Use Cases	Quick checks, reading quizzes, per-slide review	Exam prep, deep review, "connect the dots" questions

Fig 1. Comparison of Raw MCQs and KG-based MCQs (RAG)

Web Platform

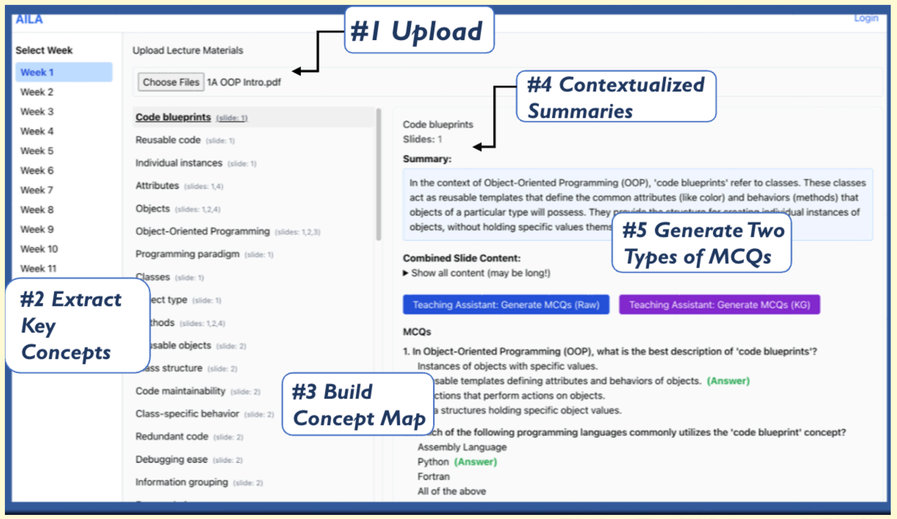

Fig 2. AILA Web System & Pipeline

Tech Stack

- ▶ **Frontend:** Next.js, React
- ▶ **Backend:** FastAPI (Python)
- ▶ **AI Model:** Gemini LLM
- ▶ **Database:** SQLite

Criterion	KG Sum	KG Avg	RAW Sum	RAW Avg	Difference (KG - RAW)
Mistakes	1285	5	1030	4.98	0.02
Hallucinations	1285	5	1032	4.99	0.01
Duplicates O	1285	5	1035	5	0
Hints	1285	5	1017	4.91	0.09
Ambiguous O	1285	5	1029	4.97	0.03
Easy Distractors	1028	4	902	4.36	-0.36
Incorrect	1285	5	1029	4.97	0.03
Ambiguous Q	1281	4.98	1026	4.96	0.02
Hard to Guess	1028	4	721	3.48	0.52
Requires Recall	1028	4	827	4	0
Correct Info	1285	5	1032	4.99	0.01
Grammatical	1285	5	1035	5	0
Correct Answer	1285	5	1035	5	0
Overall Quality	1281	4.98	1024	4.95	0.03
Overall Average		**4.783**		**4.752**	**0.031**

Fig 3. MCQ Evaluation Data

Results & Findings

▶ *KG Average: 4.78, Raw: 4.75*

MCQ Score Analysis

- **"Hard to Guess":** KG MCQs **+0.52** higher—stronger understanding.
- **"Easy Distractors":** KG MCQs **–0.36**—potential improvement area
- **Statistical Note:** Most differences, though small numerically, were consistent across >200 MCQs; "hard to guess" had a marked impact on the depth of retrieval practice.
- **Interpretation:** KG MCQs are more relevant, accurate, and integrative.

Analysis & Insights

▶ **Tradeoffs:** Raw MCQs are rapid and fine-grained aiding granular recall; KG MCQs foster synthesis, non-trivial recall supporting deeper, integrative assessment and concept integration for exam prep.

▶ **Challenges:** Variability in MCQ output, need for improved concept aggregation and instructor involvement.

▶ **Recommendations:** Use both MCQ types for balanced assessment. Refine aggregation, prompt engineering, and filtering to improve MCQ quality.

Conclusion & Future Work

▶ KG structured MCQs outperform Raw MCQs in consistency, and challenge.

▶ AILA demonstrates scalable, high-quality formative assessment generation for education.

▶ **Future:** Improve MCQ filtering, additional quality metrics, add post-use analytics, refine KG visualization, expand to other subjects.

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Github

Access full project code and documentation.